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Re:Thinking Timber Architecture. Enhancing Design and Construction Circularity through Material Digital Twin

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Keywords: Rethinking Resources, Rethinking Timber Architecture, Material Digital Twin, Material Circularity, Data Circularity, Reconfigurable Wood Architecture

Abstract Wood is a fundamental resource for building the future environment and meeting CO2 reduction goals. However, wood is truly sustainable only if it can be preserved for a long time and re-used throughout several construction cycles. Previous research by the authors shows that algorithmic design and automation technologies are introducing novel possibilities for reconfigurable architecture, which is conceived to allow easy assembly and re-assembly of wood components into different structures. In this context, dynamic and accessible material data is vital to enhance circular construction effectively. This paper introduces the concept of Material Digital Twin, an online material database directly connected to computational models and is used for real-time exchange of design, simulation, assembly, and construction information. The outlined approach is applied to the realization of a laboratory-scale prototype structure, which is used as a testbed platform to verify and document the relevance of the accessible database in supporting the automation of re-usable timber construction.

Introduction

Motivation

Circularity and renewable construction materials are necessary rather than a choice to address the climate emergency. Considering its carbon storage capacity, wood is seen as a natural asset for the future of sustainable construction. However, as the demand for a shift towards timber construction rises, the pressure on the forestal sources has increased as well. A new design paradigm that enhances circularity and re-use is urgently needed to balance the demand and supply of construction timber. In this context, computational tools and data-intensive Construction 4.0 challenge current building practices and "end-of-life" concepts by enabling materials-and-data circularity through information-driven design and construction processes.

The introduction of design-for-renewable, -reusable, -reconfigurable architecture unlocks novel properties and aesthetics at different building levels. The CREATE Group at the University of Southern Denmark has been experimenting with combinatorial and reversible design (Naboni and Kunic 2019) and robotic reassembly (Kunic et al. 2021a, b, Naboni et al. 2021) approaches to extend the life cycle of wood-based materials, components, and systems rendering them natively conceived for automated reconfiguration. Data-driven reconfigurability extends beyond a pure concept of circularity, suggesting higher design flexibility and shape transformability driven by the material's inherited performance throughout various life cycles. As part of this investigation, the *ReconWood* open project was initiated to study a series of methods and related architectural prototypes exploring the potential of circular construction through automation, particularly focusing on cyber-physical design, assembly, and disassembly of reconfigurable structures. *ReconWood* is a system based on a construction kit of discrete wood elements used to build multi-resolution framing structures that are 100% reversible, structurally optimized for reusability, and usable to fulfill the needs of a wide range of architectural and structural typologies that can evolve over time (Fig. 1). Here, *ReconBlocks* are envisioned as materials that can "travel" from one structure to another while carrying their associated datasets which allow them to be tracked, traced and reused in a data-informed fashion. Data-driven reconfigurability and re-use of building components can be effectively achieved only when material information is linked and accessible along with the material stock and shared flawlessly between different building phases.

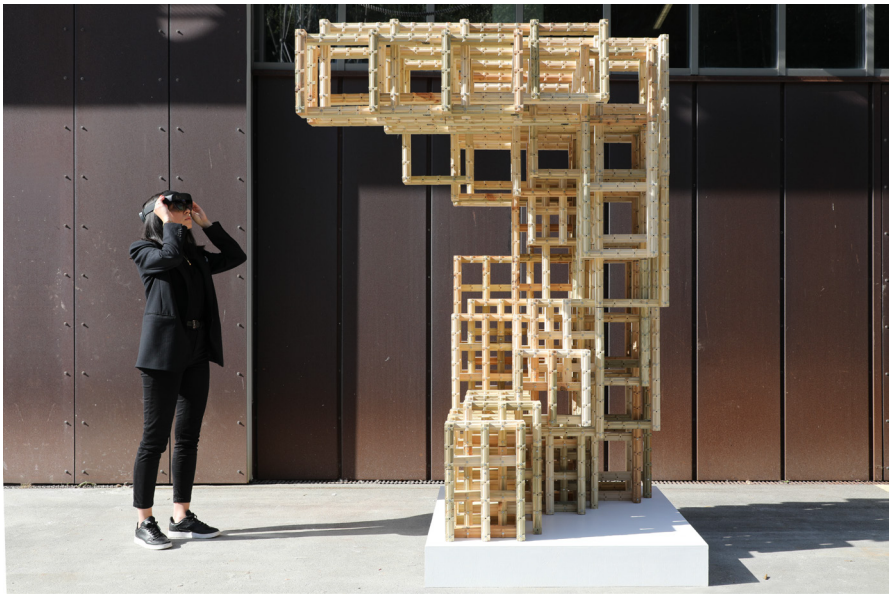


Fig 1. Interaction with the ReconWood Proto 02 in Mixed Reality - an experimental prototype for reconfigurable wood construction from the ReconWood series

Background

In recent years, research and practice-based studies aimed at reframing the construction value chain with the introduction of circularity principles. Existing buildings are now considered *material banks*, possessing a valuable material stock that can be mined for future construction up/re-cycles (Heisel and Rau-Oberhuber 2020, Honic et al. 2021). As part of the circular paradigm, a Material Passport (MP) was introduced to provide digital data sets containing information about the origin and construction history of the materials, products, and buildings and make them available for future construction uses (Heinrich and Lang 2019). Such material datasets are typically implemented at the end-of-life phase of a building, during the pre-demolition and demolition phases. Recent experimental work is instead looking into implementing MPs in the planning phase of a building, using BIM tools (Atta et al 2021, Copeland et al 2020, Jayasinghe et al 2020). On the other hand, a significant stream of research looks into reclaiming and reusing existing building materials by acquiring data from existing buildings (Wibranek et al. 2023), inspecting their maintenance conditions (Spotr, 2022), and detecting healthy and re-usable materials (Raghu et al. 2022, Akanbi 2020), and building online databases (Materiaal Marktplaats 2022, RotorDc 2015). All these approaches have, however, a common drawback: the information is statically loaded to the MP through an archive file format, permanently detaching the material information from its physical twin and design models. To enable higher levels of material circularity, more advanced data management and smoother interoperability between phases is auspicable, supported

by MPs that are “alive” platforms, constantly accessed, tracked, and updated with emerging material data (Li et al. 2021).

Research Aim

Working within the premises of Industry 4.0 (I4.0), such as interconnectivity, online data exchange, and smart automation, this research engages with a cyber-physical environment for design and construction to extend the life cycle of wood building components through data-driven reconfigurability and re-use. Data circularity is as important as material circularity to enable reconfiguration as a design and construction process. In this sense, linking material data to a material stock is fundamental. To fulfill such requirements, this research introduces the concept of Material Digital Twin (MDT), a data-driven holistic approach to support the automation of reconfigurable wood structures. In comparison to the existing concept of MP, the MDT is characterized by the following features:

1) It is established as a dynamic material database rather than a static one-shot descriptive material sheet. The database is constantly updated and incrementally loaded with new data throughout different building phases.
2) It permanently links material data to a specific building material/component using QR (quick response code) codes.
3) It is bi-directionally linked to a computational environment, from where the use-cycles are planned (design, simulation) and actuated (fabrication, assembly, disassembly).
4) It is easily accessible by both human and machine users through an internet connection.

Based on the MDT, a construction workflow is applied and tested in a laboratory setup on the design and construction of *ReconWood Proto 02*, an experimental prototype of reconfigurable wood construction where the embedded information from individual material blocks is used to assemble structurally performative aggregations. Considering the vision for these structures to be re-used, re-configured, and evolve in time, it is critical to create a watertight information loop with a shared database between different construction phases and actors.

Methodology

Material Digital Twin

The MTD is an online dynamic material database with a permanent link that enables cyber-physical interoperability and collaboration. It is used as a

middleware to bridge the virtual and physical environments and thus foster the automation of material reuse. The MDT is linked to a parametric computational model, which exchanges data from/to the following life-cycle phases: computational design, robotic fabrication, robotic assembly, use/operation, and robotic disassembly (Fig. 2).

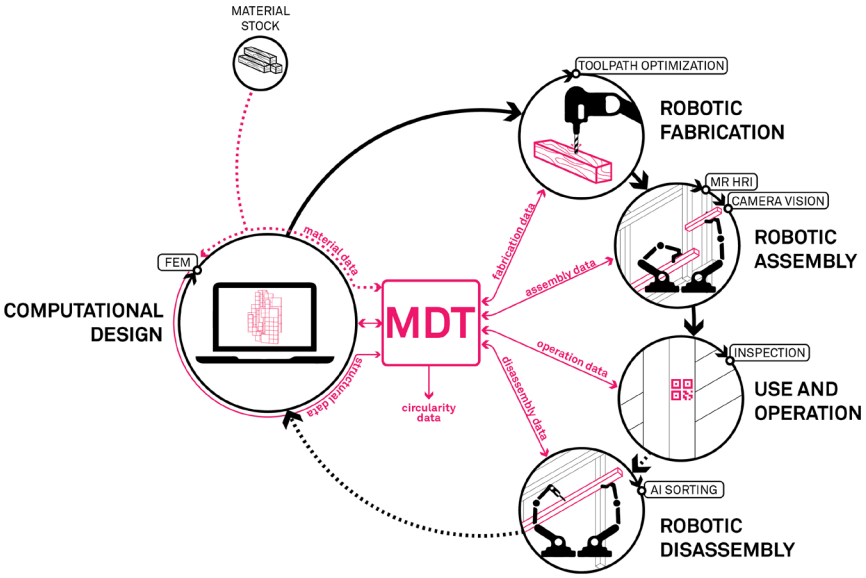


Fig 2. Schematic representation of the data exchange and communication between different construction phases and MDT

In this research, the MDT is implemented using Speckle, an interoperability application used here to extract, exchange, and visualize data in real-time (2020). A stream URL link is created and accessed through a dedicated Grasshopper 3D connector. The computational design model is developed in Grasshopper using dataset structures that refer to different construction phases, i.e., design and structural analysis, robotic fabrication, robotic assembly, use and operation, and robotic disassembly. These data are sent to the MDT and received back to Grasshopper as individual Speckle Branches, characterized by unique URL links associated with each block. QR code identifiers imprinted on the physical blocks enable the connection between physical material and its digital twin in the Speckle server. Throughout different design-construction phases, relevant datasets for circularity purposes -discussed later- are stored in the MDT to align material and data circularity within a cyber-physical loop accessible to different users, including designers, builders using Mixed Reality devices, fabrication and assembly robots which can seamlessly communicate and collaborate (Table 1).

MDT					
IDENTITY/ DESIGN	STRUCTURAL PERFORMANCE	CIRCULARITY	ROBOTIC FABRICATION	ROBOTIC ASSEMBLY	USE AND OPERATION
block ID	structural utilization	CLS (Circularity Limit State)	fabrication method	block assembly sequence No	block function in the structure
dimensions	joint mechanical performance	stored carbon	fabrication+ tools data	block assembly coordinates (x,y,z)	replacement record
cross-section	material+ characteristics	No of life cycles / uses record	fabrication+ toolpath data	block assembly orientation	material health
material data+		RR (Reusability Rate)	fabrication+ settings	holes coordinates	external treatments
material distribution		life expectancy		connections / + bolts data	environmental conditions exposure
location		future structural layer prediction		connections / + embedded nuts data	
connectivity					
reccurence in the structure					
structural layer					

Table 1 Six relevant datasets for circularity purposes sent to MDT

From the computational design phase, six datasets are extracted and loaded into the MDT. *Identity/design* shows data for each block derived from the algorithmic design choices and material stock information; *Structural Performance* data obtained from the structural simulation is used to distribute the blocks, assuring to maximize their re-usability; *Circularity* data contains predictions on the material expected performance for following cycles, combining simulation and observations on material imprecisions/decay. Datasets regarding the *Robotic Fabrication*, *Robotic Assembly*, and *Use/operation* phases are fed to MDT during the corresponding project stages, providing a rich database to inform the subsequent reconfiguration process. The Robotic Disassembly phase is not discussed here, as it is out of the scope of this paper. However, the data required for its integration and closing of the *ReconWood* circular loop is considered and can be derived from the precedent phases.

Computational Design and Circularity

The *ReconWood* construction system is based on discrete wood blocks designed to be robotically assembled and (re)assembled. These blocks are computationally aggregated into multi-resolution frame structures of variable size, shape, density, and strength. Computationally, multi-resolution layouts are obtained from a voxel-based discretization process, providing a targeted form. Resolution gradients are used globally to closely approximate the volumetric characteristics of input forms and locally as a means of structural strengthening through densification.

The *ReconWood Proto 02* is designed as a prototype structure integrating sheltering and seating areas. Here, Mixed Reality (MR) is used to start the design

process with human inputs to the form-finding process (Kunic and Naboni 2022). Subsequently, a multi-resolution frame structure is generated from a three-level voxelization layout, resulting in a 12-block construction kit. In parallel, a structural optimization algorithm is run to evaluate the structural utilization of individual timber blocks and suggest a reconfiguration to maximize their reusability. The optimization process is developed by establishing a bi-directional link between the Grasshopper computational design environment and SOFiSTiK-based numerical analysis. Specific material and joinery characteristic values, such as stiffness and strength derived from mechanical testing and loading boundary conditions, are fed to the linear FEA calculation. As a result, structural utilization values for all the blocks and activated joints are extracted. Since the goal of the design process is that each block can be reused in the future, a *Circularity Limit State* (CLS) is defined and used as a critical design optimization parameter. The available material stock consisting of white pine, pressure-impregnated spruce, and LVL wood was assigned to the blocks considering the stress distribution. In the first design iteration of *ReconWood Proto 02,* 11.2% of the structure exceeded the CLS (1), with a 204% maximum utilization value (2.04 CLS). After the optimization process, the structural density of the over-stressed area was increased to fit within the CLS, achieving a 12% utilization (0.12 CLS) (Fig. 3).

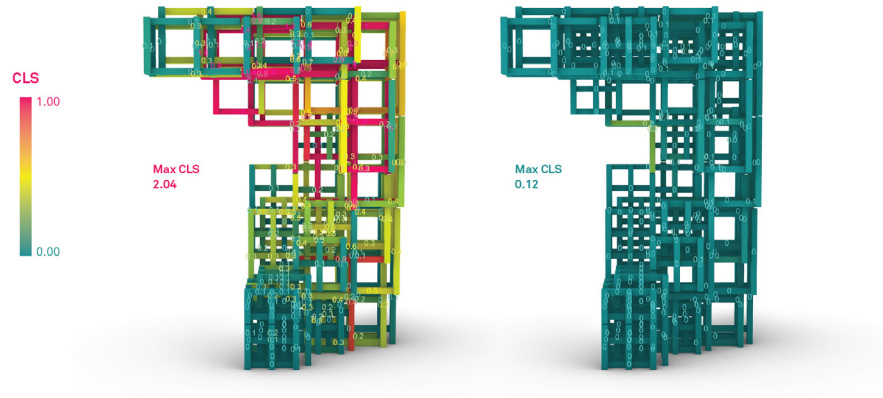


Fig 3. Linear FEA outcomes of the ReconWood Proto 02 before (left) and after (right) the optimization

Through a unique identification tag, each block inherits estimated stress values for future evaluations of fatigue and the structural strength in following construction cycles (e.g., V1B4.7, where V1 identifies the voxel level 1, B4 defines the block typology 4 and 7 is a specific count identifier). To each identified block, related data are attached and fed to the MDT through organized datasets (Table 2).

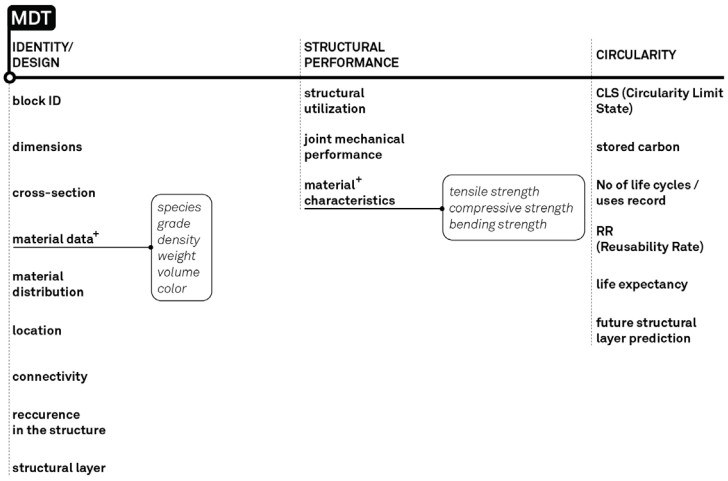


Table 2 Detailed overview of the three MDT datasets derived from the Design and Structural Simulation

Material information such as species, grade, density, weight, and volume is received from the material stock sheets, while specific structural characteristics within the structure, such as compressive, tensile, and flexural stress, are gained from simulation. On the other hand, circularity data are interpreted and predicted based on existing datasets. For instance, the *Reusability Rate* (RR) is calculated as an inverse ratio between the number of life cycles (NLC) and the CLS (1/(NLC*CL)). An individual block's life expectancy can be estimated based on simulated stresses across the various use cycles as a combination of fatigue and decay, which are stored within the MDT and made available for further elaboration.

Robotic Fabrication and Material Identification

The *ReconWood* construction kit is robotically fabricated with a mass-customization approach (Fig. 4) from three different wood types, namely LVL spruce, white pine, and pressure-impregnated spruce wood to simulate real-life conditions where the *ReconWood* blocks could be sourced from the disassembly of different constructions. The fabrication process involved using three different end tools with their corresponding toolpaths. The machining settings were adjusted depending on the wood type to ensure qualitative cuts and finishing. These parameters have been optimized through iterative milling tests and finally loaded to the MDT as *ReconWood* blocks *fabrication datasets* and made accessible for future re-fabrication purposes (Table 3).



Fig 4. Robotic fabrication of a ReconWood Block

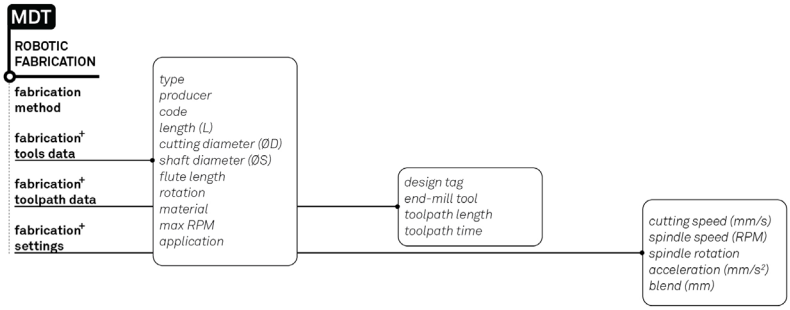


Table 3 Detailed overview of the MDT datasets derived from the robotic fabrication

During the fabrication process, the produced blocks were given an identity through a unique QR imprint, which links to their datasets in the MDT. The QR code was selected over the other tracking systems (barcode, NFC, RFID) due to the easy readability from different devices (smartphones, tablets, MR headsets, robots equipped with vision systems, etc.) and thus its predisposition for automation workflows. After comparing different techniques with hands-on tests, 35x35 mm QR codes were applied with laser cut stencil and black spray paint (Fig. 5). Numerous scanning tests confirmed that applied QR codes are easily readable by both smartphones and MR HoloLens 2 devices. Even though this method is relatively time-consuming, it provides a cost-effective and permanent transfer of QR codes to wooden blocks, which can be easily automated in an industrial context.

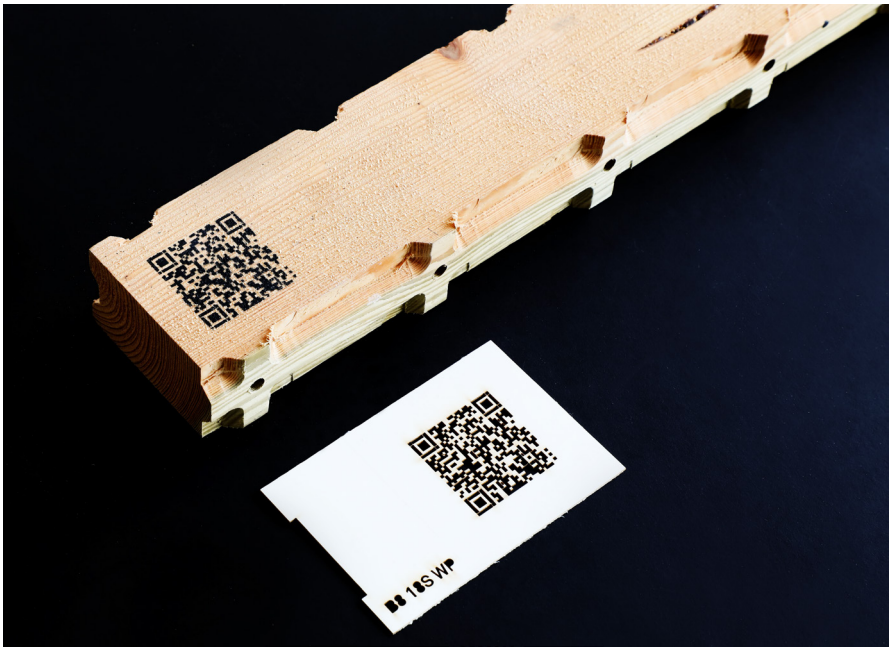


Fig 5. ReconWood block with the applied QR code and the laser-cut stencil

MR-aided Human-Robot Assembly

The readability of the QR code imprints by different digital devices allows various machines to access the MDT datasets. This supports the automation of assembly and disassembly processes and contributes to a more data-fluid cyber-physical construction environment. A previously established robotic setup for cyber-physical re-assembly of reconfigurable wood structures (Kunic et al. 2021) is here enriched by MR-aided human assistance to provide (i) an instant QR reading and identification of the *ReconWood* blocks, (ii) critical decision-making and cognitive skills during unexpected construction circumstances, and (iii) assist the robots by facilitating construction tasks that would require high-complexity and time-consuming programming and facilities.

Cyber-physical Setup - The *ReconWood* assembly setup consists of three main areas (Fig. 6). The *Assembly area* is shared between an MR-aided builder and two robots. It contains an assembly table with auxiliary materials and tools such as a bolts holder, picking/placing reference holders, and a hand screwdriver. Holographic construction data and instructions are projected in different spots to enable smooth interactivity and assistance to the builder. The *Material station* contains ReconWood blocks sorted by type, material, and holographic identifier tags. Since the station is mobile, an ArUco marker enables its tracking and referencing in space.

Finally, a *PC station* connected to Wi-Fi consists of two PCs, one running Windows with Grasshopper computational model and the other running Linux operating system with ROS. The Grasshopper computational model is the primary node that communicates bi-directionally with (1) Hololens 2 headset, (2) ROS on the PC2, and (3) MDT in Speckle.

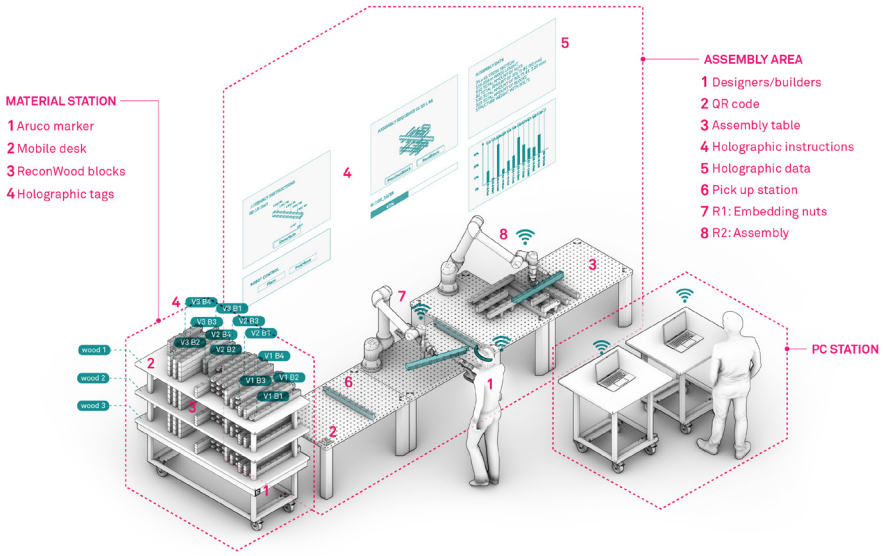


Fig 6. Cyber-physical human-robot collaborative setup for the robotic assembly/disassembly of ReconWood structures

Assembly process - The assembly instructions, such as block sequence and coordinates, and bolts and nuts locations, are communicated to the robots from the computational design model through ROS bridge connection (Kunic et al. 2021) and to the builder through a custom-made MR User Interface (UI) (Kunic and Naboni 2022). The same information is loaded to the MDT as the assembly data tree.

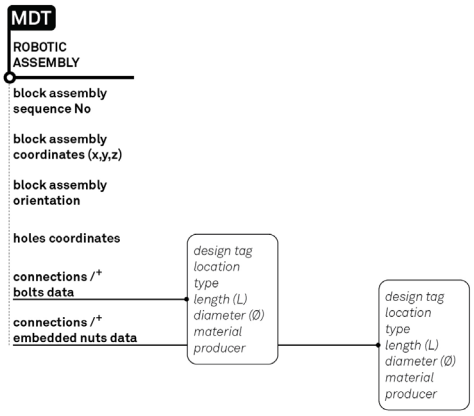


Table 4 Detailed overview of the MDT datasets derived from the human-robot collaborative assembly

Upon receiving the assembly sequence information, the builder picks the required block from the *Material station*. In the first-time assembly, the sequence of the blocks in the computational model is simply incremental (e.g., the first block type V2B4 that is met has the identification number 1). Once the builder selects a specific block (e.g., V2B4 in LVL), they scan the QR code and reads the specific identification tag. This information is then sent to the computational model and MDT to update the related design and assembly data. This operation is significant for three reasons: (1) each physical block inherits its performance data from the simulation considering its position in the structure (e.g., CLS structural data), (2) during the disassembly phase, the robots will be "aware" of the structural performance of each block and will be able to sort them accordingly, and (3) in the second-time assembly, the sequence will be defined more precisely, based on the inherited performance data from the first use of the blocks. In the following assembly step, the builder and robot 1 collaborate on the block preparation, embedding steel insert nuts into specific holes (Fig. 7). While robot 1 holds the block, the builder visualizes the nuts locations holographically and performs the operation. After this, robot 1 passes the block to robot 2 (Fig. 8), which completes the assembly process following a sequence of operations (block placement, holes detection and coordinates correction, screwing) described in Kunic et al. 2021.

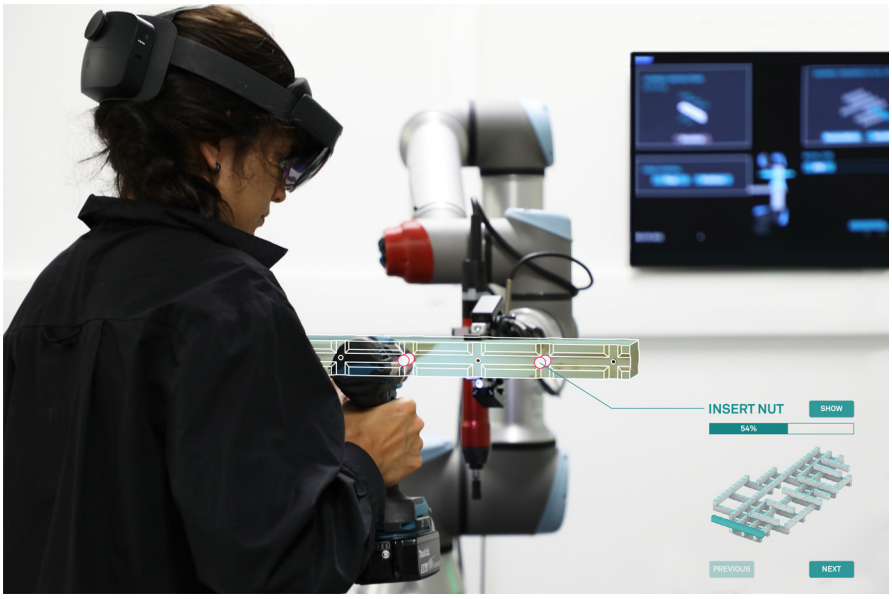


Fig 7. Human-robot collaborative process of embedding nuts

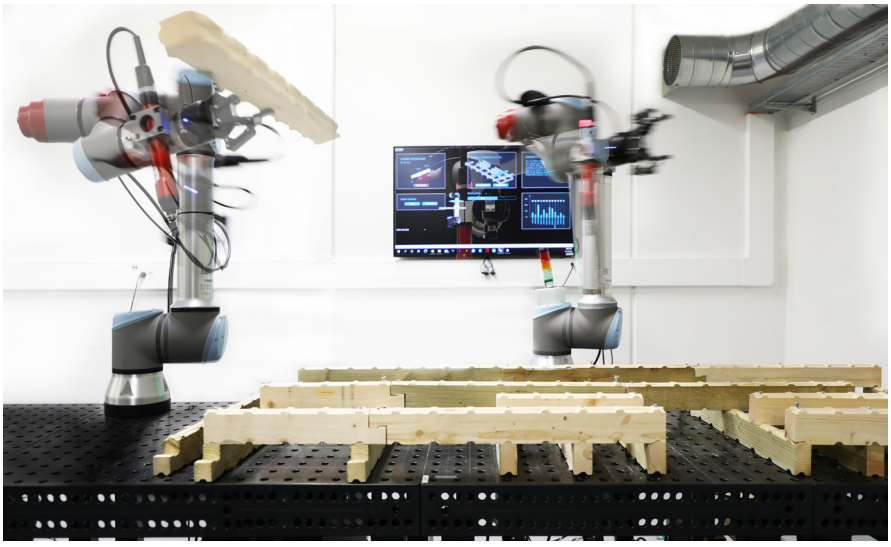


Fig 8. Multi-robot assembly of the ReconWood Proto 02

Use and Operation - Once assembled, the structure was installed upon a wooden base and exposed to the outdoor environment (Fig. 9). Additionally, the structure was spray-coated with 1-5% citric acid fire retardant and oil-based water/fungus protective paint. The used substances are recorded through an MR device to the MDT *operation data*.



Fig 9. ReconWood Proto 02 installed outside of the CREATE Laboratory at the University of Southern Denmark

Results

Design and Construction - The *ReconWood Proto 02* comprises 250 *ReconWood blocks* with a 125 m wood length of three different types and 612 reversible steel connectors. 23.2% of the structure is covered in white pine, 52.8% in impregnated spruce, and 24% in LVL. Besides meeting the structural requirements for reuse, the variety of species provides a color-rich gradient throughout the structure and introduces a new level of aesthetic detail. The prototype was successfully designed and constructed in a cyber-physical environment with human-robot collaboration, aided by MR. Each physical block carries data regarding its origin, structural performance, construction, and circularity aspects, embedded through QR code imprints and stored in the MDT (Fig. 10). All the blocks have been used for the first time in this prototype, and they are ready for full re-usability in future use cycles.



Fig 10. Close-up of the ReconWood Proto 02 showing applied QR codes on the blocks

MDT - *The ReconWood Proto 02 MDT* database holds the structured data trees regarding each building element in the form of a Speckle online link (SPeckle Stream). These are nested into 12 Speckle Branches (block type) and 250 sub-branches (unique identifier), each with six additional data layers (data categories). To facilitate the interaction with the MDT of the blocks, the six data layers are translated into visual boards in Grasshopper and sent to corresponding Speckle Branches (Fig. 11). These data can be visualized and navigated by scanning QR codes on the physical blocks. Furthermore, iFrames of the selected Speckle branches (one per block typology with six categories) are integrated into the official projects' website page to provide easy access and interactivity with the MDT datasets to a larger audience. Even though Speckle was initially conceived as a tool that enables interoperability and collaboration between different users and software, it provided a solid base for this prototype exploration of MDT and its functionalities.

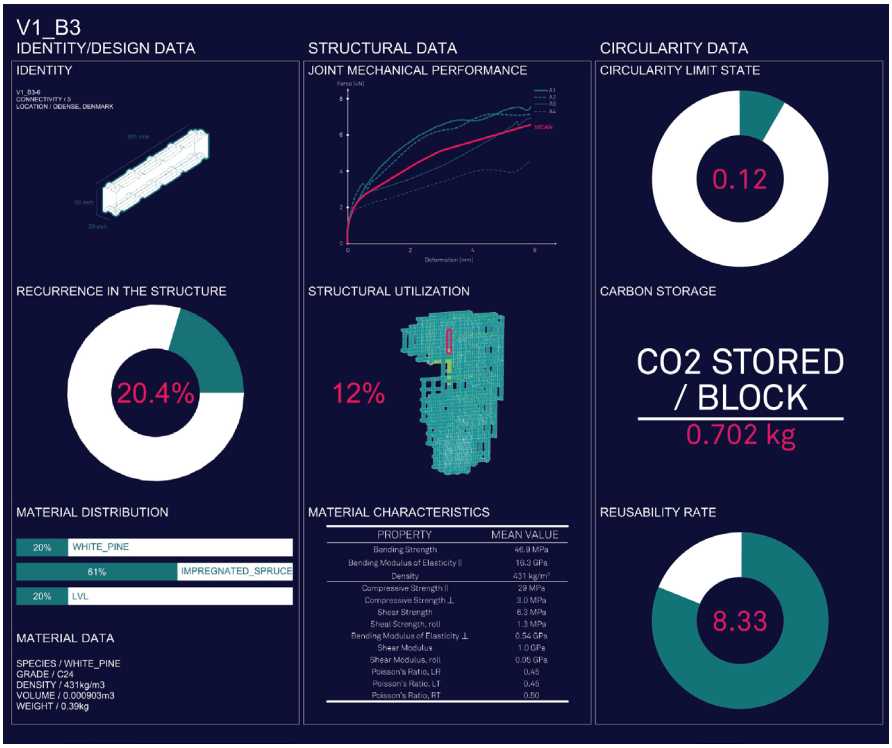


Fig 11. Example of the MDT datasets and data visualization in Speckle regarding the Identity/Design, Structural Performance, and Circularity information of the block V1_B3

Accessibility - The MDT was accessed and updated several times throughout the design-to-construction workflow. Firstly, it received the basic material stock data coming from the producer. Secondly, the designer and structural engineers collaborated on the computational design, mechanical testing, and structural optimization algorithms, where rich datasets were generated and sent to MDT. From the digital fabrication point on, the MDT could be accessed by the "machine users" since a scannable QR code has been applied to the physical *ReconWood* blocks. During the robotic assembly phase, MDT received several assembly data updates on the actual sequence of the blocks from the MR-aided human operator and the updated blocks and holes coordinates through the parametric design model, which controlled the real-time adaptive assembly operations of robot 2. Subsequently, during the final installation of the prototype, material data was updated again with specifics about the fire retardant and water/fungus protective coating. Finally, lab users and visitors had easy access to the MDT data by scanning individual QR codes on their smartphones, tablets, or MR headset devices.

Conclusions

Accessible, real-time information updates throughout the material's life cycle(s) is crucial to inform data-driven processes of circular construction. The presented prototype of MDT shows initial working functionalities that effectively support a circularity practice through the automation of design, fabrication, and re-construction phases. From a longer-term perspective, the MDT will be developed as an online platform that combines features - currently developed in Speckle - for collaboration and data sharing with a more accessible user interface for data visualization and interaction.

Project Links

Speckle link: https://speckle.xyz/streams/e25cf79b4b
Website link: https://www.create-sdu.com/projects/reconwood02

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